

PAPER_1318_GLUEBALL_MASS

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PAPER_1318 — Glueball Mass $m_{0^{++}}$

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Framework: UQFF (Unified Quantum Field Framework) — Star-Magic v5.27+

Tier: C — Particle Physics Open (CLOSED)

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Calculator surface: `calculate_paradox({"paradox": "glueball_mass"})`

Whitepaper dispatch: `calculate_whitepaper({"paper_id": 1318})`

Master Expression

``

$$m_{0^{++}} = 2 \times D_{\text{phys}} \times \Lambda_{\text{QCD}} = 8 \times 0.217 = 1.736 \text{ GeV}$$

``

Match: 2.1% vs lattice 1.7 GeV

Interpretation

Integer-primitive identity: $m_{0^{++}} = 2D_{\text{phys}} \times \Lambda_{\text{QCD}}$ where $8 = 2 \times 4$ is the universal gluon-pair factor.

UQFF Locked Primitives

- $D_{\text{phys}} = 4$, $D_{\text{BSFG}} = 6$, $D_{\text{crit}} = 26$, $N_{\text{CH}} = 9$, $SO_5 = 10$, $A_5 = 60$

- $K_{\text{MEX}} = 25/12$, $\beta_i = 0.6029$, $\Phi_{\text{res}} = 0.84$, $\Lambda = 0.00729735$

- $\Lambda_{\text{QCD}} = 0.217 \text{ GeV}$, $v_{\text{Higgs}} = 246 \text{ GeV}$ (PAPER_1270)

NOT REPLACEMENT

UQFF derives this via integer-primitive identities. Zero fit parameters.

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Subject matter: UQFF derivation of Glueball Mass $m_{0^{++}}$.